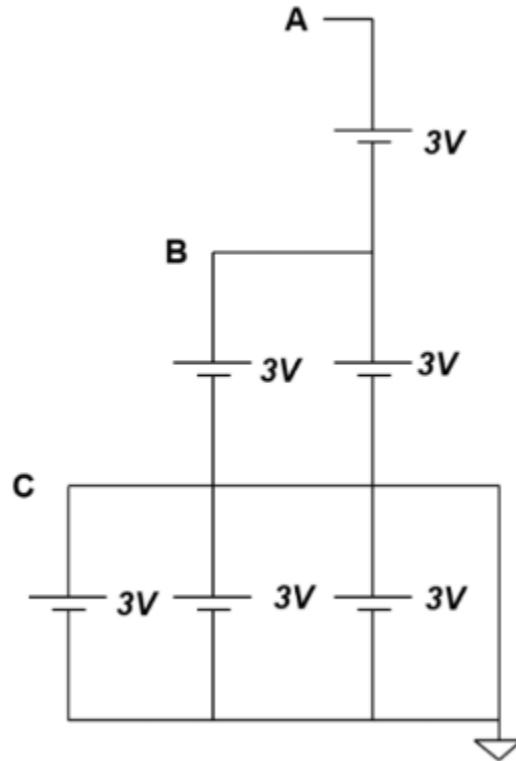


Quiz 1 Example Questions

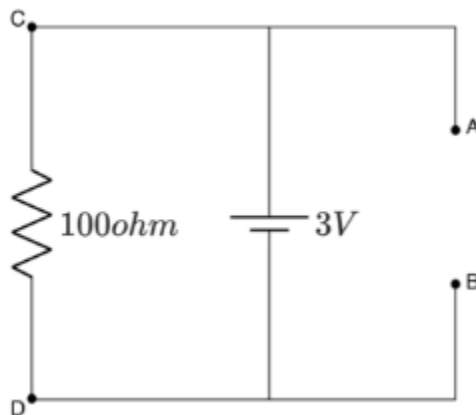
Question 1

Given the circuit diagram with nodes A, B, and C, each connected to 3V sources as shown. Determine the voltages V_A , V_B , and V_C at the respective nodes with respect to the ground.



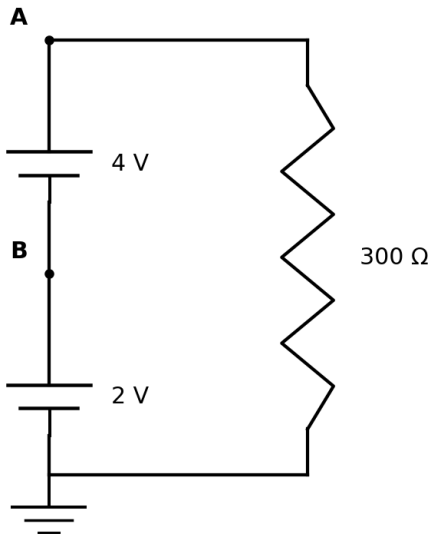
Question 2

Given the circuit with nodes A, B, C, and D as shown. Calculate the following: V_{AB} , I_{AB} , and I_{CD} .



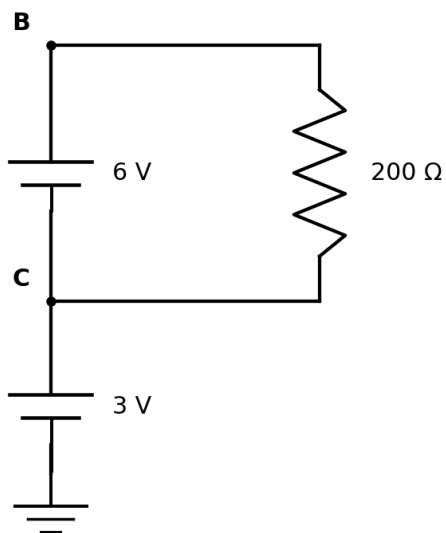
Question 3

Two ideal voltage sources are stacked in series as shown. A $300\ \Omega$ resistor is connected from node A to ground. Determine V_B , V_A , and the resistor current I_R (defined from A to ground).



Question 4

The 3 V and 6 V sources are stacked in series as shown. A $200\ \Omega$ resistor is connected between nodes B and C. Determine V_C , V_B , V_{BC} , and I_{BC} (defined from B to C).



Solutions

Question 1 Solution

- Node C is one 3 V source above ground, so $V_C = 3 \text{ V}$.
- Node B is another 3 V above node C, so $V_B = 3 \text{ V} + 3 \text{ V} = 6 \text{ V}$.
- Node A is another 3 V above node B, so $V_A = 6 \text{ V} + 3 \text{ V} = 9 \text{ V}$.

Answer: $V_A = 9 \text{ V}$, $V_B = 6 \text{ V}$, $V_C = 3 \text{ V}$.

Question 2 Solution

- A and C are connected by wire, and B and D are connected by wire. The 3 V source therefore makes the top node 3 V above the bottom node: $V_{AB} = 3 \text{ V}$.
- The A–B branch is open, so no current can flow in that branch: $I_{AB} = 0 \text{ A}$.
- The 100Ω resistor is directly across 3 V. By Ohm's law, $I_{CD} = 3 \text{ V} / 100 \Omega = 0.03 \text{ A} = 30 \text{ mA}$.

Answer: $V_{AB} = 3 \text{ V}$, $I_{AB} = 0 \text{ A}$, $I_{CD} = 30 \text{ mA}$ (from C to D).

Question 3 Solution

- The first source raises node B by 2 V above ground: $V_B = 2 \text{ V}$.
- The 4 V source is stacked on top of the 2 V source, so $V_A = 2 \text{ V} + 4 \text{ V} = 6 \text{ V}$.
- The 300Ω resistor is connected from A to ground, so it has 6 V across it. By Ohm's law, $I_R = 6 \text{ V} / 300 \Omega = 0.02 \text{ A} = 20 \text{ mA}$.

Answer: $V_B = 2 \text{ V}$, $V_A = 6 \text{ V}$, $I_R = 20 \text{ mA}$ (from A to ground).

Question 4 Solution

- The 3 V source places node C at $V_C = 3 \text{ V}$.
- The 6 V source is stacked above it, so $V_B = 3 \text{ V} + 6 \text{ V} = 9 \text{ V}$.
- The voltage from B to C is therefore $V_{BC} = 9 \text{ V} - 3 \text{ V} = 6 \text{ V}$.
- The 200Ω resistor has 6 V across it. By Ohm's law, $I_{BC} = 6 \text{ V} / 200 \Omega = 0.03 \text{ A} = 30 \text{ mA}$.

Answer: $V_C = 3 \text{ V}$, $V_B = 9 \text{ V}$, $V_{BC} = 6 \text{ V}$, $I_{BC} = 30 \text{ mA}$ (from B to C).